The Inverse of the Integer Knapsack Problem

RICE UNIVERSITY

Shammas Ahmed, Fatemeh Nosrat, Andrew J. Schaefer
Department of Computational Applied Mathematics and Operations Research

Overview

Given a (forward) optimization problem P and a feasible solution $\boldsymbol{x}^\circ$, the **inverse feasible region** is the set of objective vectors under which $\boldsymbol{x}^\circ$ is optimal to P. The inverse optimization problem finds an objective vector from this region that is closest to the original objective vector under some norm.

Applications: Radiotherapy, medical imaging [1, 2].

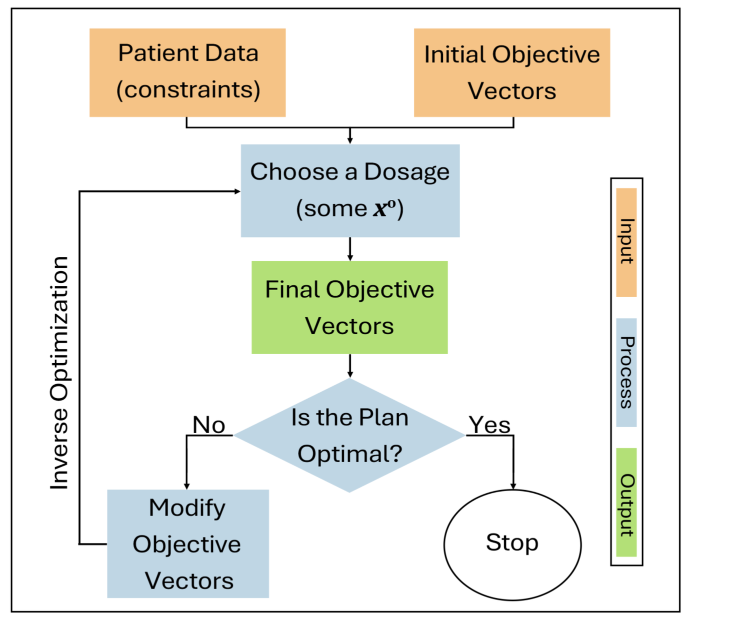


Fig 1: Inverse Optimization Framework for Radiotherapy [1].

Integer Knapsack Problem

Suppose $n \in \mathbb{Z}_{++}$ different objects, each with a profit $c_j \in \mathbb{R}$ and weight $w_j \in \mathbb{Z}_{++}, j = 1, \dots, n$, and a knapsack with a weight limit $W \in \mathbb{Z}_{++}$. The integer knapsack problem (IKP) finds the optimal quantity of every item that **maximizes** total profit and **maintains** the weight limit.

$$\text{IKP} := \max\{\boldsymbol{c}^T\boldsymbol{x} \mid \boldsymbol{w}^T\boldsymbol{x} \le W, \boldsymbol{x} \in \mathbb{Z}_+^n\}.$$

The linear programming (LP) relaxation of IKP, denoted as LP_IKP, where the integrality of decision variables is relaxed, is defined below.

$$\text{LP\_IKP} := \max\{\boldsymbol{c}^T\boldsymbol{x} \mid \boldsymbol{w}^T\boldsymbol{x} \le W, \boldsymbol{x} \in \mathbb{R}_+^n\}.$$

Inverse of Integer Knapsack Problem

The inverse feasible region of IKP with respect to $\boldsymbol{x}^\circ$, denoted as $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$, is defined below.

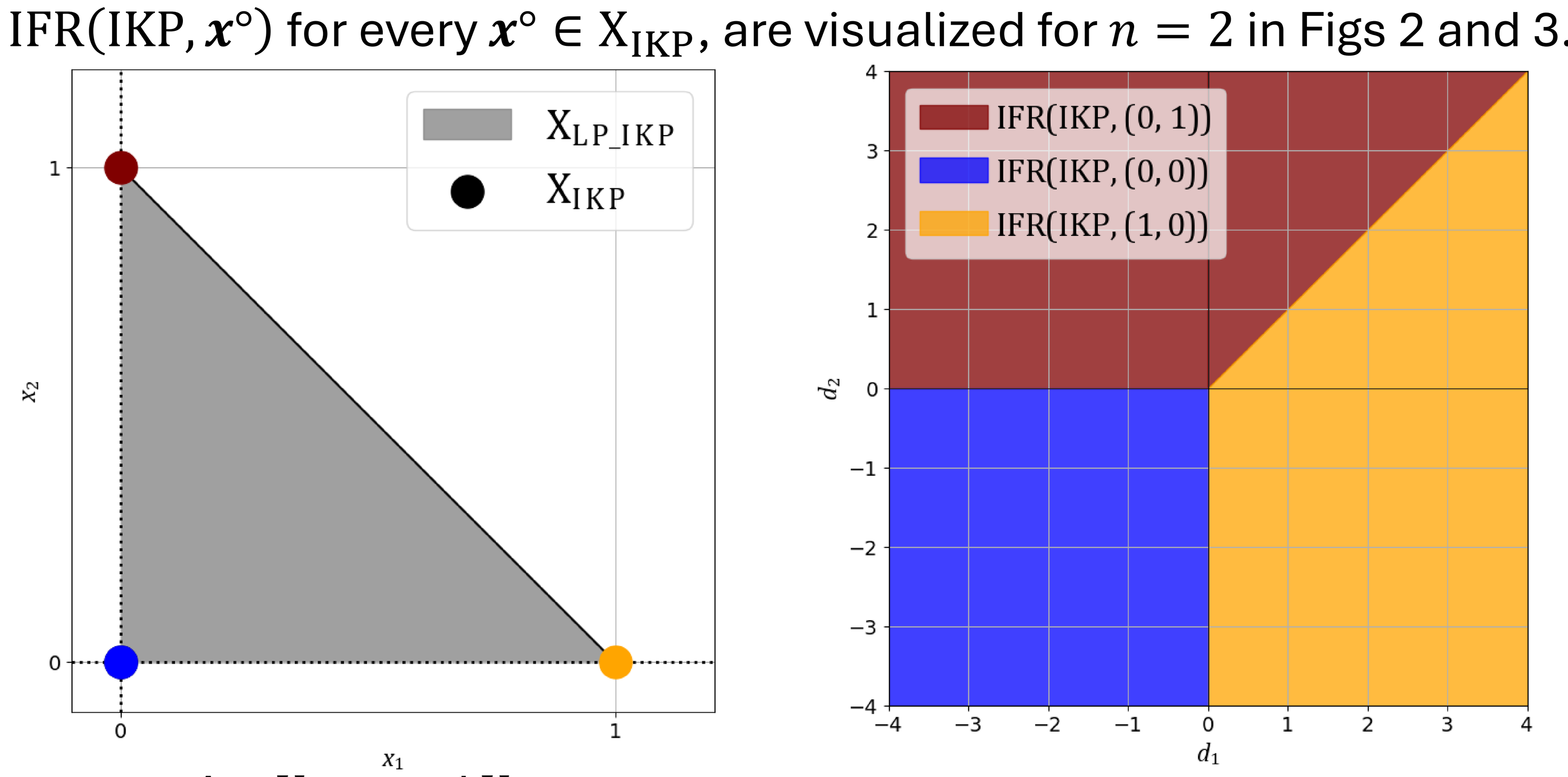
$$\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) := \left\{\boldsymbol{d} \in \mathbb{R}^n \mid \boldsymbol{d}^T\boldsymbol{x}^\circ = \max\{\boldsymbol{d}^T\boldsymbol{x} \mid \boldsymbol{w}^T\boldsymbol{x} \le W, \boldsymbol{x} \in \mathbb{Z}_+^n\}\right\}.$$

The inverse problem of IKP with respect to $\boldsymbol{x}^\circ$, denoted as $\text{IKP}^{-1}(\boldsymbol{x}^\circ)$, is defined below.

$$\text{IKP}^{-1}(\boldsymbol{x}^\circ) := \min\{\|\boldsymbol{d} - \boldsymbol{c}\|_1 \mid \boldsymbol{d} \in \text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)\}.$$

Visualization of Feasible Regions

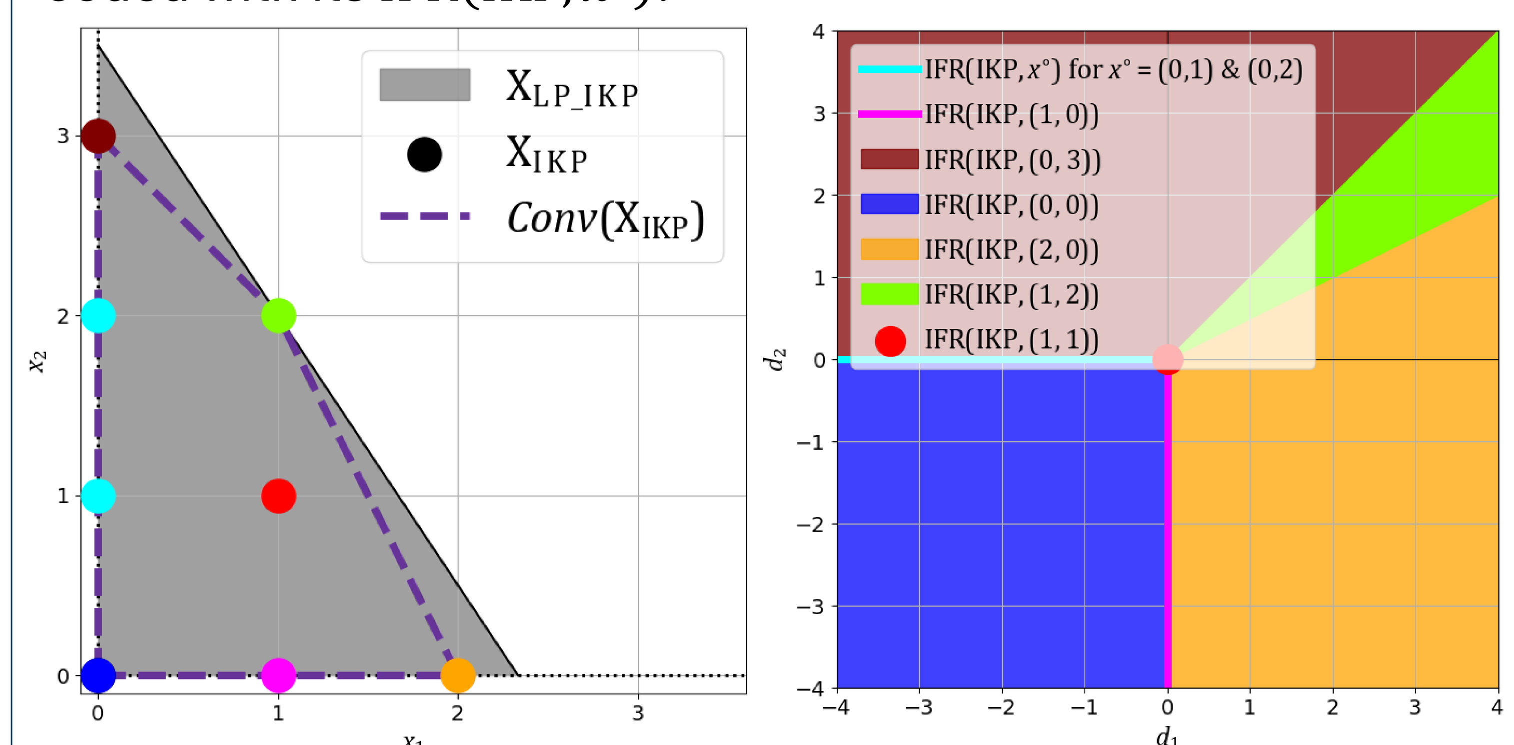
Feasible regions of IKP and LP_IKP, denoted by X_{IKP} and $\text{X}_{\text{LP\_IKP}}$, and $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$ for every $\boldsymbol{x}^\circ \in \text{X}_{\text{IKP}}$, are visualized for $n = 2$ in Figs 2 and 3.



A : X_{IKP} and $\text{X}_{\text{LP\_IKP}}$ B : $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$

Fig 2: Feasible regions of IKP, LP_IKP, and IKP^{-1} when $\boldsymbol{w} = (1, 1)$ and $W = 1$.

In many cases of LP_IKP, **boundary** and **interior points** that are feasible for IKP are present. Not all **extreme points** of LP_IKP, denoted by E, are feasible for IKP. Every $\boldsymbol{x}^\circ$ is color coded with its $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$.



A : X_{IKP} and $\text{X}_{\text{LP\_IKP}}$ B : $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$

Fig 3: Feasible regions of IKP, LP_IKP, and IKP^{-1} when $\boldsymbol{w} = (3, 2)$ and $W = 7$.

Characterization of Feasible Regions

$\text{X}_{\text{LP\_IKP}}$:
- Always bounded;
- $n + 1$ feasible extreme solutions, i.e., $|E| = n + 1$;
- $\boldsymbol{0} \in E$ is always feasible.

X_{IKP} :
- Finite;
- Not all elements in E may be feasible;
- $\boldsymbol{0} \in E$ is always feasible.

$\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$:
- Not contained in another $\text{IFR}(\text{IKP}, \boldsymbol{x})$ when $\boldsymbol{x}^\circ$ is an extreme point of $Conv(\text{X}_{\text{IKP}})$;
- Contained in another $\text{IFR}(\text{IKP}, \boldsymbol{x})$ when $\boldsymbol{x}^\circ$ is a boundary point of $Conv(\text{X}_{\text{IKP}})$;
- $\{\boldsymbol{0}\}$ when $\boldsymbol{x}^\circ$ is an interior point of $Conv(\text{X}_{\text{IKP}})$;
- Negative orthant of $\mathbb{R}^n$ when $\boldsymbol{x}^\circ = \boldsymbol{0}$.

Main Results

Assumption: All elements of E, the set of extreme solutions of LP_IKP, are feasible for IKP.

When x° is an extreme solution of LP_IKP:

1. $\bigcup_{x^\circ \in E} \text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \mathbb{R}^n$.
2. $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \{\boldsymbol{d} \in \mathbb{R}^n \mid \boldsymbol{d}^T\boldsymbol{x}^\circ = \max\{\boldsymbol{d}^T\boldsymbol{x} \mid \boldsymbol{x} \in \text{E}\}\}$.
3. If $\boldsymbol{x}^\circ = \boldsymbol{0}$, $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \{\boldsymbol{d} \in \mathbb{R}^n \mid d_j \le 0, \text{j} = 1, \dots, \text{n}\}$.
4. Let $\frac{W}{w_j} \in \mathbb{Z}_{++}$. If $\boldsymbol{x}^\circ = \frac{W}{w_j}\boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is the standard j-th basis vector of $\mathbb{R}^n$, for $j = 1, \dots, n$, then

$$\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \left\{\boldsymbol{d} \in \mathbb{R}^n \,\middle|\, d_1 \ge d_j \frac{w_1}{w_j}, \dots, d_j \ge 0, \dots, d_n \ge d_j \frac{w_n}{w_j}\right\}.$$

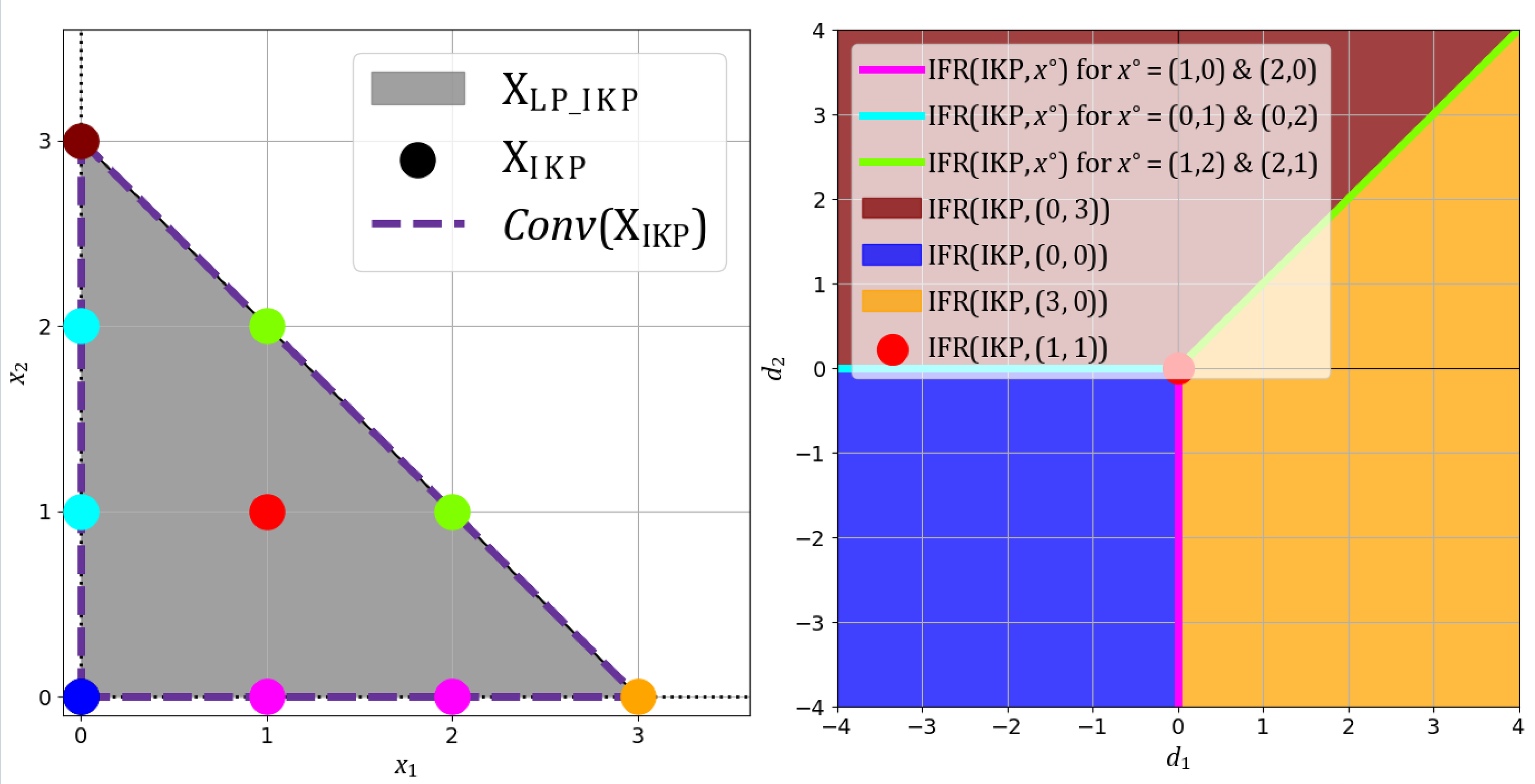
When x° is a boundary or interior solution of LP_IKP:

There exists $S_{x^\circ} \subseteq E$ such that

$$\boldsymbol{x}^\circ = \sum_{i=1}^{|S_{x^\circ}|} \lambda_i \boldsymbol{x}_i,\ \boldsymbol{x}_i \in S_{x^\circ},\ \sum_{i=1}^{|S_{x^\circ}|} \lambda_i = 1,\ \lambda_i > 0,\ i = 1, \dots, |S_{x^\circ}|.$$

1. $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \bigcap_{\boldsymbol{x} \in S_{x^\circ}} \text{IFR}(\text{IKP}, \boldsymbol{x})$.
2. If $\boldsymbol{x}^\circ$ is an interior solution of LP_IKP, then $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ) = \{\boldsymbol{0}\}$.

Visualization of Results



A : X_{IKP} and $\text{X}_{\text{LP\_IKP}}$ B : $\text{IFR}(\text{IKP}, \boldsymbol{x}^\circ)$

Fig 4: Feasible regions of IKP, LP_IKP, and IKP^{-1} when $\boldsymbol{w} = (1, 1)$ and $W = 3$.

Future Work & Acknowledgement

Future work involves (i) investigating cases where not all extreme solutions of LP_IKP are feasible for IKP and defining inverse feasible regions, (ii) finding tractable linear formulations for inverse of IKP and {0,1}-knapsack problem, and (iii) looking at the superadditive dual of IKP.

I acknowledge my mentor, **Dr. Fatemeh Nosrat** of the CMOR department at Rice University, for her continued support and guidance.

References

[1] F. Ahmadi, T. R. Mcnutt, and K. Ghobadi. Improving Observed Decisions Quality using Inverse Optimization: A Radiation Therapy Treatment Planning Application. *Arxiv*, 2024.

[2] R. K. Ahuja and J. B. Orlin. Inverse Optimization. *Operations Research*, 49(5): 771-783, 2001.

[3] A. G. Beged-Dov. Lower and Upper Bounds for the Number of Lattice Points in a Simplex. *SIAM Journal on Applied Mathematics*, 22(1): 106–108, 1972.